

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16$$

$$5^2 = 25$$

$$6^2 = 36$$

$$7^2 = 49$$

$$8^2 = 64$$

$$9^2 = 81$$

$$10^2 = 100$$

$$11^2 = 121$$

$$12^2 = 144$$

$$13^2 = 169$$

$$14^2 = 196$$

$$15^2 = 225$$

$$16^2 = 256$$

$$17^2 = 289$$

$$18^2 = 324$$

$$19^2 = 361$$

$$20^2 = 400$$

$$21^2 = 441$$

$$22^2 = 484$$

$$23^2 = 529$$

$$24^2 = 576$$

$$25^2 = 625$$

$$26^2 = 676$$

$$27^2 = 729$$

$$28^2 = 784$$

$$29^2 = 841$$

$$30^2 = 900$$

Формули за съкратено умножение:

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$a^2 - b^2 = (a+b)(a-b)$$

$$(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$$

$$(+). (+) = (-). (-) = (+)$$

$$(+): (+) = (-): (-) = (+)$$

$$(+). (-) = (-). (+) = (-)$$

$$(+): (-) = (-): (+) = (-)$$

Действия с рационални числа:

$$\frac{a}{b} \mp \frac{c}{d} = \frac{a \cdot d \mp c \cdot b}{b \cdot d}$$

$$\frac{a}{b} \mp \frac{c}{b} = \frac{a \mp c}{b}$$

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}$$

$$\frac{a}{b} : \frac{c}{d} = \frac{a \cdot d}{b \cdot c}$$

Свойства при степенуване:

$$a^m \cdot a^n = a^{m+n}$$

$$a^m : a^n = a^{m-n}$$

$$(a^m)^n = a^{m \cdot n}$$

$$(a \cdot b)^m = a^m \cdot b^m$$

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

$$\frac{1}{a^m} = a^{-m}$$

$$a^0 = 1$$

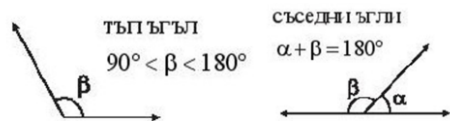
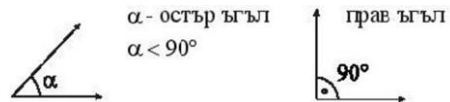
Случайно събитие:

$$\text{Вероятност на случайно събитие} = \frac{\text{Брой благоприятни изходи}}{\text{Брой на всички възможни изходи}}$$

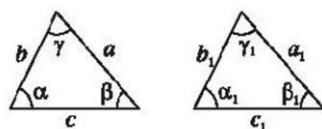
Мерни единици:

Наименование	Символ	Десетичен ножител
мили	m	10^{-3}
санти	c	10^{-2}
деци	d	10^{-1}
кило	k	10^3
мега	M	10^6

Видове ъгли:



Признаци за еднаквост:



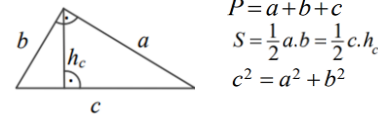
I	II	III	IV
$c = c_1$	$c = c_1$	$a = a_1$	$b = b_1$
$b = b_1$	$\alpha = \alpha_1$	$b = b_1$	$c = c_1$
$\alpha = \alpha_1$	$\beta = \beta_1$	$c = c_1$	$\gamma = \gamma_1$
			$c > b, c_1 > b_1$

За произволен
триъгълник:

$$P = a + b + c$$

$$S = \frac{1}{2} c \cdot h_c = \frac{1}{2} a \cdot h_a = \frac{1}{2} b \cdot h_b$$

Правоъгълен триъгълник:

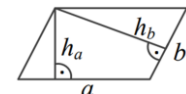


$$P = a + b + c$$

$$S = \frac{1}{2} a \cdot b = \frac{1}{2} c \cdot h_c$$

$$c^2 = a^2 + b^2$$

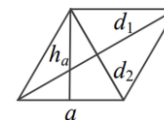
Успоредник:



$$P = 2a + 2b = 2(a + b)$$

$$S = a \cdot h_a = b \cdot h_b$$

Ромб:

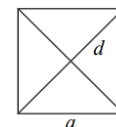


$$P = 4a$$

$$S = a \cdot h$$

$$S = \frac{1}{2} d_1 \cdot d_2$$

Квадрат:

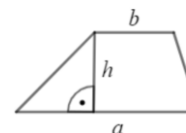


$$P = 4a$$

$$S = a^2$$

$$S = \frac{1}{2} d^2$$

Трапец:



$$S = \frac{a+b}{2} \cdot h$$

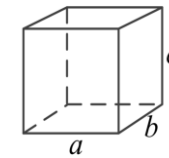
Окръжност, кръг:



$$C = 2 \cdot \pi \cdot r$$

$$S = \pi \cdot r^2$$

Правоъгълен паралелепипед:

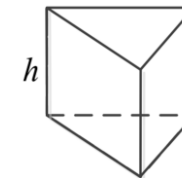


$$S = 2c(a + b)$$

$$S_1 = 2(ab + bc + ca)$$

$$V = a \cdot b \cdot c$$

Правна призма:



$$S = P \cdot h$$

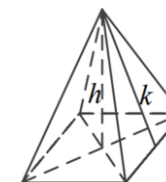
$$S_1 = S + 2 \cdot B$$

$$V = B \cdot h$$

P – обиколка на основата

B – лице на основата

Правилна пирамида:



$$S = \frac{P \cdot k}{2}$$

$$S_1 = S + B$$

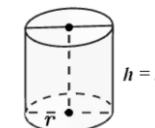
$$V = \frac{1}{3} \cdot B \cdot h$$

k – апотема

P – обиколка на основата

B – лице на основата

Прав кръгов цилиндър:

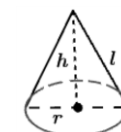


$$S = 2 \cdot \pi \cdot r \cdot l$$

$$S_1 = 2 \cdot \pi \cdot r \cdot (l + r)$$

$$V = \pi \cdot r^2 \cdot h$$

Прав кръгов конус:



$$S = \pi \cdot r \cdot l$$

$$S_1 = S + B = \pi \cdot r \cdot (l + r)$$

$$V = \frac{\pi \cdot r^2 \cdot h}{3}$$

l – образувача

Сфера и кълбо:



$$S = 4 \cdot \pi \cdot r^2$$

$$V = \frac{4}{3} \cdot \pi \cdot r^3$$